

# Case Study: ICERs

## ! Important

Please note that you can download PDF and Microsoft Word versions of this case study using the links on the right.

## Case 1 Description

You seek to explore different drug strategies to treat HIV infection. Single agent therapy with retrocyclovir at 200mg per day has a mean discounted lifetime cost of \$20,000 and results in a quality-adjusted life expectancy of 70 years. Single agent therapy with retrocyclovir at 400mg per day has a mean discounted lifetime cost of \$22,000 and a quality adjusted life expectancy of 65 years (given the increased incidence of side effects with the higher dose). Another drug, centroflexavir (which comes only in one dose of 250 mg/day) has a mean total lifetime cost of \$10,000 and a quality-adjusted life expectancy of 50 years. The combination of centroflexavir with low-dose retrocyclovir has a mean-life time cost of \$40,000 and an average quality adjusted life expectancy of 80 years, while the combination of centroflexavir with high-dose retrocyclovir has a mean lifetime cost of \$35,000 and a quality-adjusted life expectancy of 75 years.

Newer ARTs are also available and their lifetime costs/QALYs are detailed below:

Table 1: Cost and QALY Outcomes for ARTs

|         | Cost   | QALY  |
|---------|--------|-------|
| ART 1/2 | 11,000 | 63.4  |
| ART 1   | 15,525 | 73.52 |

## Instructions

*Perform a cost-effectiveness analysis of the treatment options using QALYs as the measure of health effects. Which strategy would you choose and why, if the willingness-to-pay threshold is \$50,000 per quality-adjusted life year?*

| Treatment          | Costs  | QALYs |
|--------------------|--------|-------|
| Do nothing         | 36,000 | 43.8  |
| ART 1              | 15,525 | 73.52 |
| ART 1/2            | 11,000 | 63.4  |
| Retro 200          | 20,000 | 70.0  |
| Retro 400          | 22,000 | 65.0  |
| Centro             | 10,000 | 50.0  |
| Centro / Retro 200 | 40,000 | 80.0  |
| Centro / Retro 400 | 35,000 | 75.0  |

Table 3: Treatment Options

| Strategy | Discounted<br>Costs | Discounted<br>QALYs | Incremental<br>Costs | Incremental<br>QALYs | ICER<br>(\$/QALY) |
|----------|---------------------|---------------------|----------------------|----------------------|-------------------|
|----------|---------------------|---------------------|----------------------|----------------------|-------------------|

Table 4: ICER Calculation Table 1

| Strategy | Discounted<br>Costs | Discounted<br>QALYs | Incremental<br>Costs | Incremental<br>QALYs | ICER<br>(\$/QALY) |
|----------|---------------------|---------------------|----------------------|----------------------|-------------------|
|----------|---------------------|---------------------|----------------------|----------------------|-------------------|

| Strategy | Discounted<br>Costs | Discounted<br>QALYs | Incremental<br>Costs | Incremental<br>QALYs | ICER<br>(\$/QALY) |
|----------|---------------------|---------------------|----------------------|----------------------|-------------------|
|----------|---------------------|---------------------|----------------------|----------------------|-------------------|

Table 5: ICER Calculation Table 2

| Strategy | Discounted<br>Costs | Discounted<br>QALYs | Incremental<br>Costs | Incremental<br>QALYs | ICER<br>(\$/QALY) |
|----------|---------------------|---------------------|----------------------|----------------------|-------------------|
|----------|---------------------|---------------------|----------------------|----------------------|-------------------|

## Case 2 Description

You are interested in performing a cost-effectiveness analysis of different cancer screening recommendations for anal squamous intraepithelial lesions and anal cancer in men.

Below is a table of the costs and quality-adjusted life years gained associated with different screening strategies over a six-year time span.

## Instructions

*Calculate incremental cost-effectiveness ratios for each strategy and determine which screening strategy you would choose if, as a decision maker, you were prepared to pay \$50,000/QALY (using discounted quality-adjusted life years)*

Table 6: Cost and QALY Outcomes, by Strategy

| Screening Strategy | Discounted QALYs | Discounted Costs (\$2022) |
|--------------------|------------------|---------------------------|
| No screening       | 10.05            | 4,130                     |
| Every 3 years      | 10.798           | 5,178                     |
| Every 2 years      | 10.830           | 5,583                     |
| Every 1 year       | 10.973           | 6,417                     |
| Every 6 months     | 10.988           | 8,744                     |

| Screening Strategy | Discounted QALYs | Discounted Costs (\$2022) |
|--------------------|------------------|---------------------------|
|--------------------|------------------|---------------------------|

Table 7: ICER Calculation Table 1

| Strategy | Discounted Costs | Discounted QALYs | Incremental Costs | Incremental QALYs | ICER (\$/QALY) |
|----------|------------------|------------------|-------------------|-------------------|----------------|
|----------|------------------|------------------|-------------------|-------------------|----------------|

Table 8: ICER Calculation Table 2

| Strategy | Discounted Costs | Discounted QALYs | Incremental Costs | Incremental QALYs | ICER (\$/QALY) |
|----------|------------------|------------------|-------------------|-------------------|----------------|
|----------|------------------|------------------|-------------------|-------------------|----------------|